Machine Learning Pipeline for Malignant Breast Cancer Cell Prediction

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Abstract

Aim: This study aims to develop and evaluate a machine learning pipeline for the accurate binary classification of breast tumors, using morphological features extracted from fine needle aspirate images.

Methods: Several classifiers, including Logistic Regression, Support Vector Machines (SVM), Classification and Regression Trees (CART), k-Nearest Neighbors (k-NN), and Naïve Bayes, were trained on the Wisconsin Diagnostic Breast Cancer dataset. The pipeline included thorough preprocessing, feature selection methods (ANOVA F-test and Recursive Feature Elimination), hyperparameter optimization via GridSearchCV, and model evaluation using stratified 10-fold cross-validation.

Results: Logistic Regression achieved the best overall performance, with an accuracy of 98.2%, an AUC of 0.998, and an MCC of 0.963. CART also showed strong performance (96.5% accuracy), offering higher interpretability. Feature importance analysis highlighted worst perimeter, worst area, and worst concavity as the most relevant predictors of malignancy, aligning with known clinical indicators.

Conclusion: The proposed pipeline effectively combines high predictive performance with clinical interpretability, demonstrating significant potential for practical application.

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Supplementary information: Supplementary data are available at *giacomo-timelli* online on GitHub.

1 Introduction

Breast cancer represents one of the most critical global public health challenges, being the most frequently diagnosed cancer among women and a leading cause of cancer-related mortality. Early and accurate diagnosis plays a fundamental role in improving treatment outcomes and increasing survival rates. In recent years, the integration of computational approaches into medical diagnostics has opened new perspectives in cancer detection, with machine learning emerging as a powerful tool for pattern recognition and classification in biomedical applications.

1.1 Biological Background

The morphological features of cell nuclei obtained through fine needle aspiration (FNA) biopsies provide essential diagnostic information for breast cancer detection. Malignant cells typically present distinct characteristics compared to benign ones, including larger nuclear size, irregular shape, less smooth contours, and reduced symmetry.

These morphological differences can be quantitatively described using a set of computed features, such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset captures these properties through digital image analysis, offering a comprehensive set of features suitable for computational modeling.

Each feature is expressed through three statistical measures: mean, standard error, and "worst" (defined as the mean of the three largest values). This structure provides a detailed numerical representation of nuclear morphology, forming the foundation for automated classification approaches.

1.2 Machine Learning in Cancer Diagnostics

Machine learning techniques have significantly advanced the field of cancer diagnostics by enabling the development of predictive models capable of identifying complex patterns within high-dimensional biomedical data. In particular, supervised learning algorithms, especially classification models, can learn the relationship between nuclear morphological features and tumor malignancy using labeled datasets.

Once trained, these models can generalize the learned patterns to accurately classify new, unseen cases, supporting clinical decision-making processes.

1.3 Study Objectives

The objective of this study is to design and implement a comprehensive machine learning pipeline for the binary classification of breast tumors using the WDBC dataset. The ultimate goal is to identify the most effective predictive model for breast cancer classification, with potential applications in clinical decision support systems.

2 Materials

The dataset employed in this study is the Breast Cancer Wisconsin Diagnostic Dataset, which is publicly available through the UCI Machine Learning Repository. It consists of 569 samples, each corresponding to a breast mass, including 357 benign cases (class 0) and 212 malignant cases (class 1).

Each sample is characterized by 30 real-valued features, obtained from three statistical measures—mean, standard error, and "worst" (defined as the mean of the three largest values)—computed for ten different nuclear characteristics. These features provide a comprehensive quantitative description of cell morphology, enabling effective computational analysis and classification.

2.1 Features description

The dataset includes ten key features used to characterize the morphological properties of cell nuclei. Each feature captures a specific aspect of nuclear shape or texture, contributing to the overall classification process:

- **Radius**: Mean of the distances from the center to points on the perimeter.
- **Texture**: Standard deviation of gray-scale values.
- **Perimeter**: Length of the nucleus contour.
- **Area**: Area enclosed by the contour.
- **Smoothness**: Local variation in radius lengths.
- **Compactness**: Defined as (Perimeter² / Area) − 1.0.
- **Concavity**: Severity of concave portions of the contour.
- **Concave Points**: Number of concave regions along the contour.
- **Symmetry**: Degree of symmetry of the nucleus shape.
- **Fractal Dimension**: Approximation of the contour's fractal dimension.

2.2 Software and Libraries

The entire analytical workflow was implemented in Python within a Google Colab environment, providing a consistent and reproducible computational framework. The implementation leveraged several widely used scientific computing libraries.
Data handling and manipulation were carried out using pandas for structured data operations and **NumPy** for numerical computations.
Data visualization and figure generation were performed using **Matplotlib** and **Seaborn**, enabling the creation of static, publication-quality plots.
Machine learning task, including data preprocessing, feature selection, model training, hyperparameter tuning, and performance evaluation, were conducted using the **scikit-learn** library.
The use of these well-established open-source tools ensured methodological rigor, reproducibility, and alignment with current best practices in data science.

3 Methods

3.1 Data Preparation

The dataset was loaded into a pandas DataFrame, and all columns were properly labeled to reflect the corresponding feature names and statistical measures. A preliminary data quality check was conducted to identify missing values, confirming that the dataset was complete.
For reproducibility purposes, the dataset was then exported and stored in CSV format, ensuring that all subsequent analyses could be replicated consistently.

3.2 Exploratory Data Analysis (EDA)

An exploratory data analysis (EDA) was conducted to better understand the dataset structure, including class distribution, feature distributions, and relationships between variables. This step was essential for identifying potential issues such as imbalance, skewness, and multicollinearity before model development.

Class Distribution
The dataset shows a moderate class imbalance, with 357 benign cases and 212 malignant cases. While not extreme, this imbalance is relevant for model evaluation and motivated the use of appropriate performance metrics beyond simple accuracy (Figure 1 in additional_materials).

Feature Distributions
Density plots and boxplots were generated for all 30 features to analyze their statistical distributions and assess differences between benign and malignant classes.
These visualizations highlighted:

- Variability in feature distributions
- Presence of skewness in some variables
- Clear separation patterns for several features, suggesting strong discriminative power

In particular, features related to tumor size and shape showed noticeable differences between classes, supporting their relevance for classification tasks (Figure 2 and Figure 3 in additional_materials).

Correlation Analysis
To investigate relationships between features, a correlation heatmap based on Pearson's correlation coefficient was computed. The coefficient is defined as:

$$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2\sum(Y-\bar{Y})^2}}$$

where r ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation).
The analysis revealed several pairs of features with extremely high positive correlations ($r > 0.95$), particularly among morphologically related variables such as radius, perimeter, and area (both mean and worst values).
These strong correlations indicate the presence of **multicollinearity**, which can:

- Inflate variance in regression-based models
- Reduce model interpretability

- Introduce redundancy in the feature space

Based on these findings, feature selection techniques were later applied to remove redundant variables and improve model generalization (Figure 4 in additional_materials).

Biological Interpretation

Beyond statistical implications, the correlation structure also supports known clinical insights. Specifically, strong positive relationships among size-related features reflect the biological characteristic that malignant tumors tend to have larger and more irregular nuclei.

All visual outputs referenced in this section (class distribution plots, density plots, boxplots, and correlation heatmaps) are provided in a separate file. For a complete visual interpretation of the analysis, please refer to the **additional_materials**.

3.3 Data Rescaling and Train/Test Split

Before model training, a comprehensive preprocessing pipeline was implemented to ensure data quality and optimize algorithm performance. As observed during the exploratory data analysis (EDA), several features exhibited substantial variability in scale. For instance, variables such as *area* had values in the hundreds, whereas features like *smoothness* and *symmetry* were typically bounded between 0 and 1.

This discrepancy represents a critical issue for scale-sensitive algorithms, particularly Logistic Regression, which relies on gradient-based optimization. In such cases, features with larger numerical ranges can disproportionately influence model coefficients, potentially leading to biased learning and suboptimal convergence.

To address this issue, feature standardization was performed using the StandardScaler from Scikit-learn, which rescales each feature to have zero mean and unit variance according to:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the feature mean and σ is the standard deviation.

This transformation ensures that all variables contribute equally to the learning process, regardless of their original scale. To avoid data leakage, scaling parameters were computed exclusively on the training set and then applied to both training and test data using a fit-transform approach.

Finally, the dataset was split into training and testing subsets using a stratified strategy, allocating 80% of the data for training and 20% for testing while preserving class distribution.

3.4 Feature Selection

Feature selection was performed to improve model performance, reduce overfitting, and enhance computational efficiency. This step was particularly important given the relatively high dimensionality of the dataset (30 features) and the presence of multicollinearity identified during correlation analysis.

Two complementary approaches were adopted to identify the most informative features:

- **Univariate Feature Selection (ANOVA F-test):** The ANOVA F-test was used to evaluate the discriminative power of each feature independently. The F-score is defined as the ratio between between-group variance and within-group variance.

Features were ranked according to their F-scores, and the top k features with the highest discriminative ability were selected.

- **Recursive Feature Elimination (RFE) with Logistic Regression:** To complement the univariate approach, Recursive Feature Elimination (RFE) was applied as a multivariate method.

This wrapper technique iteratively removes less important features based on model coefficients through the following process:

1. Train a Logistic Regression model with L2 regularization on all features
2. Rank features by the absolute value of their coefficients
3. Remove the least important feature
4. Repeat until the desired number of features is reached

The Logistic Regression model used in RFE incorporated balanced class weights to account for class imbalance, ensuring that feature importance was not biased toward the majority class.

The feature selection process was validated using cross-validation to determine the optimal number of features that maximized predictive performance while minimizing redundancy. The subsets obtained from both methods were compared, with particular attention to preserving biological interpretability and clinical relevance.

Overall, this strategy effectively mitigated multicollinearity and produced a refined feature set capturing the most informative morphological characteristics.

3.5 Model Training and Evaluation

The core of the analysis involved training and comparing six supervised learning algorithms:

1. **Logistic Regression (LR):** Estimates the probability of a binary outcome using a logistic function
2. **Linear Discriminant Analysis (LDA):** Assumes normally distributed features with equal covariance matrices and projects data to maximize class separation
3. **k-Nearest Neighbors (k-NN):** Classifies samples based on the majority class among the k closest neighbors
4. **Classification and Regression Trees (CART):** Builds a decision tree by recursively splitting the data

5. **Gaussian Naive Bayes (NB):** Probabilistic classifier based on Bayes' theorem assuming feature independence
6. **Support Vector Machines (SVM):** Identifies the optimal hyperplane maximizing class separation

To ensure robustness and generalizability, a k-fold cross-validation strategy (k = 10) was employed during training. This approach partitions the data into *k* subsets and iteratively trains on *k-1* folds while validating on the remaining one, reducing overfitting and providing a more stable performance estimate.
Hyperparameter optimization was conducted using GridSearchCV from Scikit-learn, which performs an exhaustive search across predefined parameter grids (e.g., regularization strength *C*, number of neighbors *k*, tree depth). The best configuration was selected based on cross-validation performance and then evaluated on the test set.
Each optimized model was assessed using multiple complementary metrics:

- **Accuracy:** Proportion of correctly classified samples
- **Confusion Matrix:** Breakdown of true positives, true negatives, false positives, and false negatives
- **Classification Report:** Includes precision, recall, and F1-score for each class
- **ROC Curve and AUC:** Evaluate discrimination ability across thresholds
- **Matthews Correlation Coefficient (MCC):** A robust metric for imbalanced datasets, ranging from -1 to +1

All performance metrics for each model are available in the project's GitHub repository within the *eda* folder.

4 Results

The implemented machine learning pipeline enabled a robust and systematic evaluation of six classifiers, providing clear insights into their effectiveness for breast cancer diagnosis. All results were obtained using a strict hold-out test set and are summarized through a comparative analysis.

4.1 Overall Model Performance

All models achieved strong predictive performance, with accuracy values exceeding 90%, highlighting the high discriminative power of nuclear morphological features in the WDBC dataset. A summary of the key evaluation metrics is reported below:

Model	Accuracy	F1 Score	MCC	AUC
Logistic Regression	0.982	0.976	0.963	0.998
Support Vector Machine	0.974	0.963	0.944	0.995
CART	0.965	0.950	0.926	0.952
k-Nearest Neighbors	0.939	0.911	0.869	0.982
Gaussian Naïve Bayes	0.921	0.889	0.829	0.989

Table 3. Performance metrics of all classifiers evaluated on the test set.

Logistic Regression emerged as the best-performing model across all metrics. With an accuracy of 98.2%, an F1-score of 97.6%, and an AUC of 0.998, it demonstrates near-perfect discriminative capability. The high Matthews Correlation Coefficient (MCC = 0.963) further confirms robust performance, even in the presence of class imbalance.

This result can be attributed to effective regularization achieved during hyperparameter tuning (optimal *C* value identified via GridSearchCV in Scikit-learn), which allowed the model to avoid overfitting while preserving its ability to capture relevant linear relationships in the data.

The Support Vector Machine also achieved excellent performance (Accuracy: 97.4%, AUC: 0.995), closely following Logistic Regression. These results suggest the presence of some non-linear patterns; however, the slightly lower performance indicates that a linear decision boundary is likely more suitable for this dataset.

4.2 Analysis of Learning Algorithms and Biological Interpretability

The observed performance ranking (LR > SVM > CART > k-NN > NB) provides insight into the underlying structure of the classification problem.

The superior performance of Logistic Regression suggests that the separation between benign and malignant nuclei is largely linear within the selected feature space. Its effectiveness is driven by the model's ability to assign optimal weights to highly discriminative features identified during feature selection, such as *mean radius*, *mean concavity*, *worst perimeter*, and *worst area*.

The strong performance of CART (Accuracy: 96.5%, AUC: 0.952) is particularly noteworthy. Despite its simplicity compared to ensemble methods, the decision tree achieved high accuracy through careful pruning and parameter tuning (*max_depth = 5*, *min_samples_leaf = 2*). Importantly, the resulting tree provides transparent and interpretable decision rules, closely reflecting clinical reasoning processes and facilitating trust in model predictions.

4.3 Evaluation of Class Imbalance Handling

Despite the inherent class imbalance in the dataset (63% benign vs 37% malignant), all models maintained strong performance on the minority class. Logistic Regression, in particular, achieved well-balanced results, with an F1-score of 97.6% for malignant cases. This is a critical outcome, as minimizing false negatives is essential in medical diagnostics, where failing to detect a malignant tumor can have severe consequences.

The consistently high MCC values across all models, especially the 0.963 achieved by Logistic Regression, indicate that the classifiers successfully learned meaningful discriminative patterns rather than exploiting class distribution biases.

5 Discussion and conclusions

This study successfully developed and validated a robust machine learning pipeline for the binary classification of breast tumors using the Wisconsin Diagnostic Breast Cancer dataset. The analysis, covering all stages from exploratory data analysis to hyperparameter tuning and model evaluation,

provides several relevant insights for both machine learning applications and clinical diagnostics.

5.1 Summary of Key Findings

The results demonstrate that machine learning models, particularly Logistic Regression, can achieve high levels of accuracy and reliability in breast cancer classification. Logistic Regression emerged as the best-performing model, achieving 98.2% accuracy and an AUC of 0.998. Its strength lies not only in performance but also in interpretability, as it assigns clear weights to biologically meaningful features.
Although Support Vector Machines (SVM) showed comparable performance, they consistently performed slightly worse across evaluation metrics and offer lower interpretability, making them less suitable in this context. The CART model, on the other hand, provided strong results and represents a valuable alternative when transparent, rule-based decision-making is required, closely mirroring clinical reasoning processes.
Overall, these findings confirm that quantitative morphological features hold strong diagnostic power in distinguishing between benign and malignant breast tumors.

5.2 Clinical Implications and Reliability

A key outcome of this study extends beyond traditional performance metrics. The analysis revealed a strong alignment between model-derived feature importance and established clinical knowledge. Specifically, the model consistently identified *worst perimeter*, *worst area*, and *mean concavity* as the most predictive features.
This is consistent with clinical understanding, where malignant cells are characterized by larger and more irregular nuclei. Such agreement is crucial for building trust among clinicians in AI-assisted diagnostic systems, as it indicates that the model's decision-making process reflects biologically grounded reasoning rather than purely statistical patterns.

5.3 Methodological Contributions

The developed pipeline offers several methodological strengths:

- **Reproducibility:** The complete workflow—from data acquisition to model deployment—ensures that results can be reliably reproduced.
- **Comprehensive Evaluation:** The use of both threshold-dependent metrics (Accuracy, F1-score) and threshold-independent metrics (AUC, MCC) allows for a more complete assessment of model performance.
- **Interpretability vs Precision Trade-off:** The comparison between interpretable models (CART) and high-performing models (Logistic Regression) provides practical insights for selecting models based on clinical needs.

5.4 Limitations and Future Directions

Despite promising results, several limitations must be acknowledged. The primary limitation lies in the dataset itself. Although widely used, the Wisconsin Diagnostic Breast Cancer dataset is relatively small (569 samples) and derived from a single institution. This may limit the model's generalizability across diverse patient populations and clinical settings.
Additionally, the dataset may not fully represent the variability observed in real-world clinical scenarios, including rare tumor subtypes. Future research should focus on validating the model using larger, multi-institutional datasets to better assess its robustness and generalizability.
Regarding class imbalance, while class weighting proved effective, more advanced techniques could be explored. Methods such as SMOTE, ADASYN, or cost-sensitive learning may further enhance performance, particularly by improving sensitivity in detecting malignant cases.

5.5 Conclusion

In conclusion, this study shows that machine learning models—especially Logistic Regression—can function as highly accurate and clinically meaningful tools for breast cancer classification. The strong consistency between model outputs and established clinical knowledge represents a significant step toward developing trustworthy AI systems in healthcare.
The proposed pipeline provides a solid and adaptable framework for other medical classification problems, effectively balancing predictive performance with interpretability.

5.6 Supplementary Materials

To ensure transparency, reproducibility, and accessibility, all project materials—including the Jupyter notebook, processed datasets, generated visualizations, trained models, and supporting code—have been made publicly available on GitHub at the following repository: https://github.com/giacomo-timelli/AML_project_breast_cancer_classification

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